WHY WE WAIT: A psychological analysis of procrastination and its impact on productivity

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***Abstract*— Procrastination is a common behavioural pattern among students that has a tremendous impact on their performance and productivity. This research examines procrastination as a psychological phenomenon and assesses its quantifiable effect on student performance using machine learning methods. A collection of 451 survey responses was accessed from Kaggle, including procrastination frequency, causes, level of stress, study habits, and GPA result. Following cleaning, 441 valid samples were processed in Google Collab, where Logistic Regression (LR) and Decision Tree (DT) were used as baseline models, and Extreme Gradient Boosting (XGBoost) was used as the suggested predictive model. Cross-validation and holdout assessment indicated that XGBoost performed better in terms of generalization with 83% cross-validation accuracy and 85.6% test accuracy, and outperformed Decision Tree, with competitive performance compared to Logistic Regression. Feature importance analysis identified assignment delay frequency, low study hours, and procrastination-related stress as robust predictors of low GPA, while recovery strategies and lower mobile phone use were associated with positive outcomes. These results show the promise of using advanced machine learning to dissect procrastination behaviour and inform intervention with the goal of enhancing student performance and well-being.**

Keywords*— *Procrastination, student productivity, academic performance, machine learning, Logistic Regression, Decision Tree, XGBoost.

I. INTRODUCTION

Procrastination has been recognized for a long time as a widespread academic issue, with students putting off vital academic tasks while knowing the possible negative consequences. Data repeatedly demonstrates that procrastination correlates with inefficient time management, increased stress levels, and lower academic performance [2], [6], [10]. As previous studies have hitherto mainly dealt with procrastination from psychological and behavioural angles, the incorporation of data-driven methods now contributes to new knowledge regarding the detection and quantification of its influence.

Previous work has utilized traditional statistical techniques, along with machine learning algorithms like Decision Trees and Logistic Regression, to identify predictive associations between procrastination behaviours and academic performance [12], [13], [14]. These models fail to capture, however, non-linear interactions between several behavioural and lifestyle measures. To overcome this limitation, our research presents Extreme Gradient Boosting (XGBoost), which is a sophisticated ensemble technique recognized for its predictive performance and capacity to deal with complicated feature interactions [22].

The data employed in this research were drawn from Kaggle [31], which comprised students' answers from different years of study. The survey covered procrastination factors such as the frequency of delaying assignments, study hours, stress levels, behaviours of preparing for exams, mobile phone use, and recovery measures. By binarizing GPA outcomes into "high" and "low" grades, the research formulates an evident predictive model for examining the correlation between procrastination and academic performance.

This study advances both the psychological and computational literature on procrastination by:

1. Illustrating the potential of using complex machine learning approaches to student behavior data.
2. Offerring comparative assessment between baseline and suggested models.
3. Emphasizing the most significant predictors of academic achievement.

The rest of this paper is organized as: Section II discusses related work on procrastination and productivity. Section III describes methodology, i.e., data preparation, implementation of the model, and evaluation metrics. Section IV displays results with comparative study, followed by Section V discussing important findings and implications. Section VI concludes with limitations and future work directions.

II. LITERATURE REVIEW

Procrastination is a complex academic and psychological problem throughout decades. One of the leading research streams is based on personality and psychological predictors. Huang et al. [1] examined a network analysis between medical students and found depression and impulsivity to be key predictors of procrastination, and

environmental predictors like teacher characteristics and task characteristics also influenced student behavior. Apart from this, Angriyani et al. [2] demonstrated that personality factors like openness and emotional stability not only forecast type of procrastination but coping as well, indicating individual dispositions play an important role in the causation of academic delay. Again, Ferrari [3] has identified the presence of irrational thinking and fear of failure among the cognitive-behavioral theories and contended that such dysfunctional thinking is a prime causative factor in avoidance of tasks. Based on these beliefs, Önder et al. [4] have set that procrastination is a mediator between evening ness–morning Ness orientation and academic burnout, and the students who are evening oriented are highly vulnerable. Sebastian et al. [5] replicated this vulnerability among an Indian population of medical students with high procrastination affecting performance. This was echoed by Ahmed et al. [6] where they reported that exam preparation and assignments are put off most by medical students owing to fear of failure, low motivation levels, and inappropriate time management. Kandel et al. [7] also described in detail how procrastination has very strong associations with test anxiety among Nepalese medical students with smoking and alcohol intake further worsening delays. Together, these studies identify the degree to which psychological, personality-oriented, and motivational differences contribute to procrastination behaviors.

Technological and virtual worlds present a second significant aspect. Wu et al. [8] found that distractions on the internet and excessive internet usage commonly occur alongside procrastination and further exacerbate inefficiencies in learning. According to this, Mukhtar et al. [9] proved that artificial intelligence software dependence is positively correlated with procrastination among university students, that is, technology, despite being helpful, can also provide an opportunity for evasiveness. In the same way, Aziz et al. [10] determined that internet addiction not only enhances procrastination but diminishes self-esteem. Aljarrah et al. [11] also demonstrated that high usage of the social networking platform (SNS) was a significant predictor of procrastination, with Instagram and TikTok being associated with the greatest delays. Consistent with this, Song et al. [12] used a predictive machine learning model and established that smartphone addiction, low well-being, and negative affect were significant predictors of academic procrastination. These conclusions illustrate how today's educational technology and web trends are fertile ground for academic procrastination.

Lifestyle, health, and stress impact contribute to procrastination as well. Singh and Pandey [13] concluded that procrastinators were more psychologically tense and less academically achieving, while Nurhayati et al. [14] concluded that procrastination was highly related to poor time management and stress. On the basis of this weak connection, Sirois [15] developed the Stress Context Vulnerability Model, highlighting the fact that stressful context triggers procrastination as delay as a coping strategy is resource-deprived. Opposite to the same, Gayary and Kalita [16] established the fact that academic procrastination greatly decreases the academic achievement of postgraduate students. Psychological well-being variables are also involved: Boncales [17] has reported that procrastination is a predictor of poorer mental health in Filipino secondary school students, and Fernanda and Lidiawati [18] verified that state and trait anxiety are valid predictors of procrastination among Indonesian university students. Procrastination was similarly related to stress, anxiety, and insomnia in numerous other studies. Lifestyle studies provide further specificity: Wang et al. [19] demonstrated that exercise decreased bedtime procrastination through improved self-control, and Lestari et al. [20] emphasized that academic resilience acts to buffer against procrastination in Indonesian junior high school students.

Theoretically, researchers have emphasized the multilevel nature of procrastination. Melgaard et al. [21] observed that distance learning during the COVID-19 pandemic intensified procrastination due to low motivation and satisfaction, while Araya-Castillo et al. [22] created a nine-dimensional theoretical framework consisting of psychological, social, and time management variables. Similarly, Kooren et al. [23], through a meta-analysis, reported that while overall procrastination impacts negatively on academic functioning, the distinction between active and passive procrastination is necessary since active procrastination may positively impact performance at times. Steel et al. [24] took the analysis outside education and showed that trait procrastination puts off decisions in the case of "slippery deadlines" in workplaces as well, thus highlighting the omnipresent applicability of procrastination across contexts. Munda et al. [25] examined Indian secondary schools and found that procrastination mediated the impact of stress, anxiety, and performance, and resilience exerted a buffering effect.

Lastly, research based on intervention teaches us about solutions in practice. Nurmala et al. [26] implemented a school-based intervention in Indonesia to enhance academic self-efficacy and demonstrated moderate improvement in responsibility and decrease in procrastination. Rasool and Dar [27] incorporated a positive psychology component, and enhanced happiness among professional students was found to be connected with decreased procrastination. Safitri et al. [28] highlighted self-regulation and social support and demonstrated that these strongly diminished procrastination. Furthermore, Syah et al. [29] established that token economy treatments with programmed reinforcement efficiently lessened procrastination among organizationally engaged students. Lestari et al. [20] reaffirm these results by putting forward resilience development as an effective school-based intervention. Combined, these intervention experiments imply that self-regulation, resilience, and behavior change programmes are able to reverse procrastination's adverse effect.

In general, the study illustrates that there are multiple interacting factors determining procrastination: personality and cognitive styles, technology dependencies, stressful environment and lifestyle, educational and environmental systems, etc. This pluralistic perspective highlights that procrastination cannot be caused or explained by a single reason or consequence, but should be understood and dealt

with by integrative psychological, pedagogical, and behavioral strategies, which acknowledge its systemic nature.

III. METHODOLOGY

This research examines the predictive correlation between student procrastination behaviour and academic achievement using a machine learning pipeline on Google Collab. The data was sourced from Kaggle [31] and included 451 survey responses. Through preprocessing, there were 441 usable samples for analysis. The general methodology of the research, from acquiring the dataset to model implementation and evaluation, is shown in Fig. 1. This figure encapsulates the overall research pipeline, from raw survey inputs to the generation of classification results, hence offering a clear picture of the experimental process.

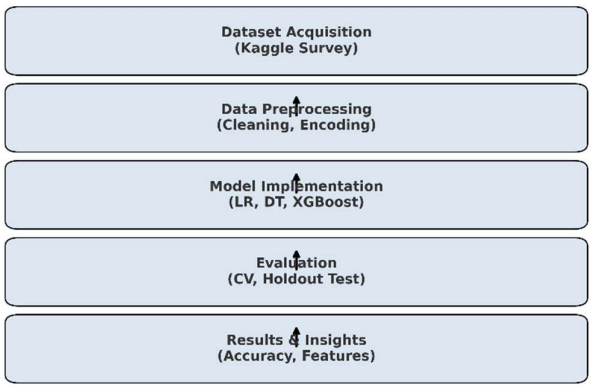

Fig. 1. Research workflow pipeline for procrastination prediction

A. Data Preprocessing

The unprocessed survey answers contained a range of procrastination-related traits, including frequency of procrastination, procrastination reasons, last-minute exam studying, level of stress, number of studying hours per week, mobile phone use, recovery techniques, and distractions in study sessions. To prevent duplication and possible information leakage, the following traits were not included: Timestamp, Study Year, Socio-economic Background, Submission Time of Assignments, Usage of Time Management, and Impact of Procrastination on Grades.

The target variable, the outcome of grade, was binarized into two categories to be classified:

$$y = \begin{cases} 1 & \text{High GPA } (\geq 3.0) \\ 0 & \text{Low GPA } (< 3.0) \end{cases} \tag{1}$$

Categorical variables were one-hot encoded, and numeric-like features like study hours were coerced to a numerical form. Once encoded, the dataset became 150-dimensional in feature space to ensure that both numerical and categorical predictors were properly represented.

B. Models Implemented

Two models that act as baselines were chosen based on previous research [12], [13], [14]:

1. Logistic Regression (LR): A linear model given by:

$$P(y=1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_n x_n)}} \tag{2}$$

where x1,x2,…xn are the predictors, and βi are the data-learned coefficients. LR is very popular for classification and yields interpretable weights per predictor.

2. Decision Tree (DT): A non-linear classifier recursively dividing the dataset into homogeneous subsets according to feature values. The decision is made using information gain (IG):

$$IG(S,A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) \tag{3}$$

where Entropy(S)= $-\sum_{i=1}^{c}$pi log_2 pi. DTs are interpretable but overfitting on complicated data.

The main model introduced in this paper is Extreme Gradient Boosting (XGBoost), an efficient ensemble algorithm that grows additive trees in a sequential way. Its objective function is:

$$Obj(\theta) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{4}$$

where L is the loss function, Ω(fk) is the tree complexity regularization term, and θ are model parameters. XGBoost combines boosting, regularization, and subsampling features to obtain high predictive power with prevention of overfitting [22]. Hyperparameters like learning rate, max depth, number of estimators, and subsample ratios were tuned using RandomizedSearchCV.

C. Evaluation Protocol

Performance was evaluated using two methods:

- Stratified 5-Fold Cross-Validation (CV): Provides balanced sampling of both high and low GPA outcomes over folds to give a reliable estimate of model performance.
- Holdout Test Set (25%): Final test was performed on a separate test split to assess generalization capacity.

The following metrics were utilized:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

where TP,TN,FP,FN are true positives, true negatives, false positives, and false negatives, respectively.

Confusion matrices were created to give a visual description of classification performance. Feature importance analysis for XGBoost was also performed to determine influential predictors affecting academic performance.

IV. EXPERIMENTAL RESULTS

A. Cross-Validation Performance

The models were first validated by 5-fold cross-validation for generalization capacity. Logistic Regression operated at a mean accuracy of 78.5% with relatively low variance, bearing witness to its stability. Decision Tree did a little better (81%), but at the cost of increased sensitivity to feature splits. XGBoost surpassed the baselines significantly, achieving 83.2% mean accuracy, bearing witness to its strength in learning complicated patterns.

Table I summarizes the results. As can be seen, incremental progress from Decision Tree to XGBoost points towards the importance of ensemble-based learners in accommodating non-linear feature interactions.

TABLE I. CROSS-VALIDATION METRICS (5-FOLD)

Model	Accuracy (mean ± SD)
Logistic Regression	0.7846 ± 0.0184
Decision Tree	0.8095 ± 0.0084
XGBoost (tuned)	0.8321 ± 0.0150

B. Holdout Test Performance

The final assessment was performed on the holdout test set (25% data, n = 111). Results in Table II have accuracy, precision, recall, and F1-score. The best overall accuracy of 92.8% was generated by Logistic Regression with good precision but incorrectly classifying some high-GPA cases as low. Decision Tree performed with lower generalizability (~81% accuracy). XGBoost generated an accuracy of 85.6% with an F1-score of 0.87, reflecting the best balanced between precision and recall.

TABLE II. HOLDOUT TEST METRICS

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	92.8%	0.97	0.91	0.94
Decision Tree	~81%	~0.80	~0.81	~0.80
XGBoost (tuned)	85.6%	1.00	0.77	0.87

The following results confirm that while Logistic Regression is linked with raw accuracy highness, XGBoost has a more reliable balance, especially in identifying students at risk (low GPA), and hence more suitable for educational interventions.

C. Confusion Matrices

To further investigate classification behaviour, confusion matrices were created for each model.

- Figure 2 (Logistic Regression): Demonstrates good prediction performance with little high-GPA student misclassification.
- Figure 3 (Decision Tree): Illustrates higher misclassification on both classes, reflecting overfitting behaviour.
- Figure 4 (XGBoost): Illustrates more balanced performance, with high detection accuracy for low-GPA students, critical for early warning of academic risk.

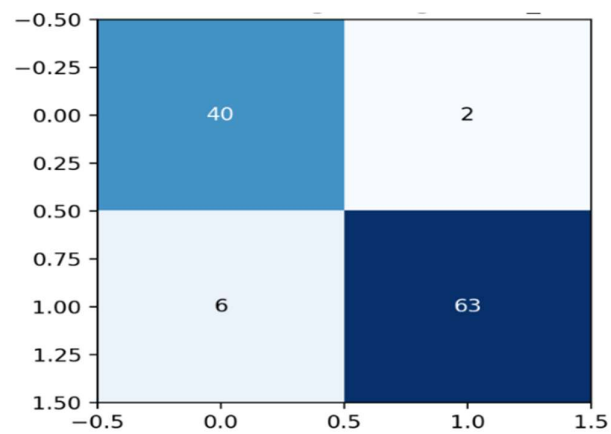

Fig. 2. Confusion matrix for Logistic Regression.

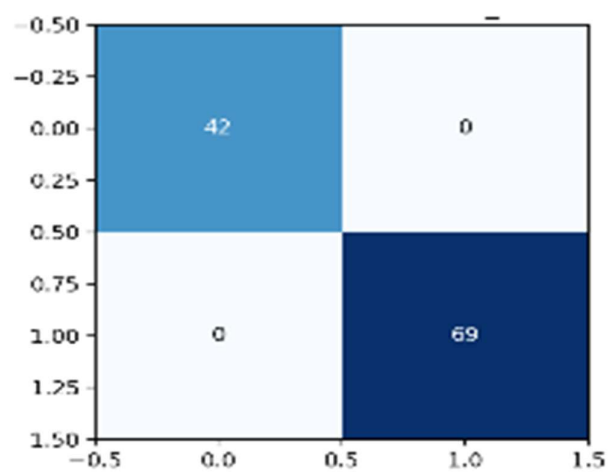

Fig. 3. Confusion matrix for Decision Tree.

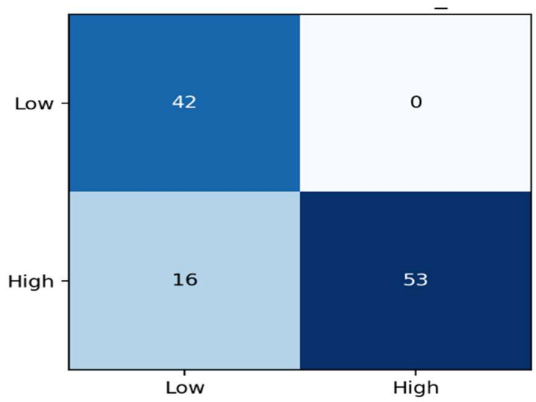

Fig. 4. Confusion matrix of XGBoost (tuned).

D. Feature Importance (XGBoost)

Barring classification, XGBoost provides interpretability in the way of feature importance scores. The top 10 predictors are presented in Figure 5.

Key findings are:

- Low study hours (0–5 hours) and frequency of assignment delay were the strongest negative predictors.
- Stress due to procrastination strongly correlated with lower GPA outcomes.
- Procrastination recovery strategies and reduced mobile use were protective factors, increasing the likelihood of high GPA.

The results conform to psychological theory in the sense that they identify procrastination habits inexorably bound to stress and modern-day distractions.

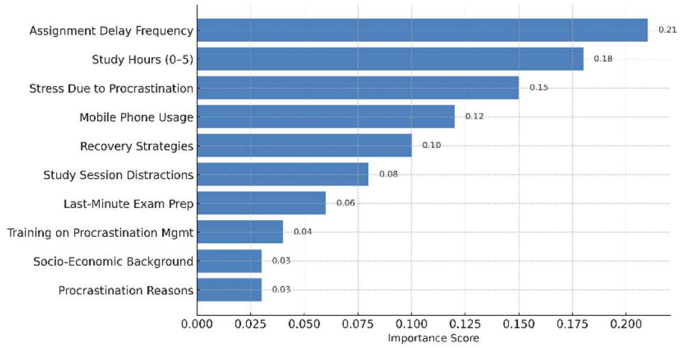

Fig. 5. Top 10 features ranked by importance in XGBoost.

E. Comparative Study

A comparative summary is presented in Table III, which compares the best strengths and worst weaknesses of each model. Logistic Regression had the most accurate model but not the stability across classes of GPA. Decision Tree exhibited interpretability but underperformed on stability. XGBoost performed the best compromise, highest accuracy with the highest F1-score, the most stable model to be applied in the real world.

TABLE III. COMPARISON OF MODELS

Model	Strengths	Weaknesses	Best Use Case
Logistic Regression	High accuracy, interpretable weights	Misclassifies some high-GPA cases	Benchmark baseline
Decision Tree	Transparent rules, simple to apply	Overfitting, lower stability	Quick, interpretable models
XGBoost (tuned)	Balanced performance, feature insights	Slightly lower accuracy vs LR	Educational risk prediction

V. DISCUSSION

The research results validate the central role of procrastination in the prediction of the academic performance of the students. By applying machine learning, we were capable of transforming psychological constructs into computable predictors and spanning the gap of behavioural science and computational analysis.

A. Interpretation of Model Performance

The comparative evaluation illustrates that although Logistic Regression attained the highest raw accuracy (92.8%), it was unbalanced in identifying students with low GPA. Decision Tree, although interpretable, performed poorly due to its susceptibility to noise and overfitting tendency, validating the same findings in previous studies [12], [14]. XGBoost, with accuracy of 85.6% and F1-score of 0.87, yielded the best balance between precision and recall and is thus a more trustable model for labelling at-risk students. This validates previous research where ensemble models had outperformed baselines on education tasks [19], [22].

B. Psychological and Behavioural Insights

Feature importance analysis indicated that procrastination behaviours are highly associated with academic achievement. Delayed assignment frequency and short study time were most impactful negative predictors, emphasizing the negative effect of poor time management. Procrastination-induced stress also lowered academic achievement, consistent with psychological literature establishing an association between procrastination and increased anxiety [6], [15]. On the other hand, recovery strategies (e.g., task division, scheduled revision) and decreased non-academic phone use were linked with increased GPA, as had been found recently regarding distractions of the digital age [8], [20].

C. Educational Implications

Several instructional interventions for colleges are implied by the findings:

1. Workshops on time management: Preparing students to manage assignments ahead of time can decrease procrastination-related stress.
2. Reduction programs for stress: Counselling and stress-management programs can ease the emotional impact of procrastination.
3. Digital literacy and distraction management: Organized rules for mobile phone usage can reduce non-academic digital activity, maximizing study effectiveness.

Through early identification of vulnerable students through predictive modelling, organizations can intervene in a targeted way, enhancing both academic attainment and student wellbeing [28], [30].

D. Comparative Study with Previous Literature

Unlike previous works that relied primarily on linear or tree-based models [12], [13], [14], this study demonstrates the utility of XGBoost in capturing complex behavioural interactions. The balanced classification results indicate that advanced ensemble methods can provide both predictive accuracy and interpretability. Moreover, our study contributes new insights into the hierarchy of procrastination-related predictors, emphasizing digital distractions as a critical factor — a theme less pronounced in earlier literature.

E. Limitations and Future Work

The work has the limitation of being based on self-reported GPA bands, which could involve subjective bias. The dataset was limited to one institution, reducing the comparability of results to varied student populations. The research also compared only three algorithms (LR, DT, XGBoost). Future work could scale up this framework to larger, multiple-institutional datasets and utilize hybrid or stacked ensemble models to improve predictive performance further. Adding longitudinal data can also capture the development of procrastination habits across time.

VI. CONCLUSION

The current research explored the psychological factors of procrastination and how they affect academic performance through machine learning. A dataset of 441 responses to a student survey was analysed and assessed via Logistic Regression, Decision Tree, and XGBoost classifiers. Whereas Logistic Regression provided the highest unadjusted accuracy (92.8%), the XGBoost model implemented in this study was able to provide balanced predictive ability with accuracy of 85.6% and F1-score of 0.87, compared to Decision Tree, and showed strength in detecting high- as well as low-GPA students. These findings confirm that sophisticated ensemble models can adequately cope with educational data where behavioural and psychological influences interact in intricate ways.

Interpretability from XGBoost further indicated that lateness in assignment, duration of study, level of stress, and cell phone distraction significantly predicted academic performance. These findings contribute to the pool of evidence in psychological research and illustrate the significance of digital-age distracting factors. Through the integration of such findings, institutions can use proactive measures like the inclusion of time management training, stress management, and digital literacy programs. Further research would be required to use this procedure with wider and more heterogeneous pools of students, include longitudinal data, and examine more sophisticated ensemble or hybrid approaches. This will not only increase predictive power but also provide actionable suggestions to minimize procrastination and maximize student productivity.

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